

the Atlantic to attack New York. Wells' vision of an air raid was apocalyptic: "As the airships sailed along they smashed up the city... Below, they left ruins and blazing conflagrations and heaped and scattered dead." But although the Germans hoped to create a Zeppelin bombing fleet, cooler heads realized that aircraft of the period were neither

reliable enough nor capable of carrying a large enough bombload to wreak such devastation on enemy cities. While experiments were carried out by dropping bombs and firing guns from airplanes – the majority of which were aimed at enemy troops – such offensive uses of aviation made relatively little progress before 1914. Instead more modest, but still vital, military roles were emphasized, centered on reconnaissance, message carrying, and artillery spotting – that is, helping gunners hit their targets by telling them where their shells were landing. By 1912, aircraft fulfilling these roles had become a standard feature of military maneuvers, and experiments had begun in air-to-ground communication (including the use of radio) and aerial photography.

Military demand

Although airships were added to the resources of both navies and armies, airplanes generally proved themselves more useful and reliable. They were also far cheaper to produce – a very important consideration. It was only in Germany that Zeppelin advocates held their ground, diverting major resources away from airplane production.

Between 1911 and 1914, European military establishments became major buyers of airplanes and the main influence on the development of the air industry. Military competitions set manufacturers targets to aim at, with lucrative contracts at stake. The lure of profits brought substantial investment from the likes of German banker Hugo Stinnes, arms manufacturer Gustav Krupps, and Russian industrialist Mikhail Shidlovski. Some private firms experienced rapid growth. Henri Farman (see page 32) was employing around 1,000 workers by 1914, and the Gnome aero-engine company operated on a similar scale. Governments also set up their own establishments to encourage aircraft development – notably Britain's Royal Aircraft Factory at Farnborough.

However, the situation in the United States was strikingly different. America was not preparing for a major war. Its armed forces were under no pressure

to embrace cutting-edge technology, and its politicians were reluctant to allocate funds for military hardware. By the summer of 1913, when the biggest European military air arms were numbered in hundreds, the US Army had 15 airplanes. Without substantial military contracts, the American air industry stagnated. In 1914, only 168 Americans were employed making aircraft.

Aircraft designs

The domination of European aviation by military contracts brought a distinct change in priorities. Since they did not yet take seriously the prospect of combat in the air, the armed forces demanded

sturdy, reliable aircraft that could be flown in most weather conditions by average pilots and still carry a reasonable payload. Sporting pilots willingly risked their lives in treacherous high-performance machines built for speed or for stunting, but the military wanted stable airplanes that would survive prolonged use and keep their newly trained pilots alive. Although light monoplanes continued to be ordered for army use – for example, Taubes in Germany and Morane-Saulniers in France – there was a strong prejudice in favor of solid biplanes. A typical example was the two-seater B.E.2, designed by Geoffrey de Havilland for the Royal Aircraft Factory in 1912.



FLYING SOLDIERS
German military officers – among the first of such to take to the sky – pose in flying gear beside a Harlan Eindecker monoplane during an instruction course (c.1910–13).

FIRST BOMBING RAID

IN THE FALL OF 1911, Italy declared war with Turkey in a dispute over the territory now known as Libya, then part of the decaying Ottoman Empire. The Italian army possessed a number of foreign aircraft – French Blériots, Farmans, and Nieuports, and German Taubes. An air flotilla, initially comprising just nine airplanes and 11 pilots, was sent off with the Italian force that embarked for the Libyan coast in North Africa. In the short but brutal war that followed, the airplanes performed creditably, carrying out reconnaissance missions, mapping areas of the desert, and dropping propaganda leaflets promising a gold coin and sack of wheat to all those who surrendered. On November 1, Lieutenant

Giulio Gavotti dropped four grenades over the side of his Blériot on to an Ottoman military encampment at the Taguira oasis, in the first ever bombing raid by an airplane. Despite the fact that they faced little opposition, the aviators were hailed as heroes by patriotic Italians. Although the 1899 Hague Convention banned aerial bombing from balloons, Italy argued that this ban could not be extended to airplanes.



BATTLE OF DERNA
This propaganda poster shows three Italian monoplanes circling the battle of Derna during the Italo-Turkish war (1911–12).